Bangzheng Sun

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Research Interests

- **High-redshift galaxy evolution**: galaxy populations at $z \geq 6$; evolved galaxies at $z \geq 4$
- **Time-domain science**: high-redshift transients ($z \geq 3$); variable AGNs
- **Wide-field surveys**: imaging and spectroscopy with JWST, Roman, and Euclid

Academic Appointments

California Institute of Technology 2026 - Now
Infrared Processing and Analysis Center (IPAC) Postdoctoral Scholar
· *Supervisor: Dr. Yun Wang*

Education

University of Missouri - Columbia 2020 - 2026
Ph.D. in Astronomy
· *Thesis:* New Insights into Galaxy Evolution - A Quartet of JWST Studies
· *Advisor:* Dr. Haojing Yan

University of Wisconsin - Madison 2016 - 2020
B.Sc. in Astrophysics, Physics, & Mathematics
· *Thesis:* X-Ray Spectra and Light Curves of Interacting White Dwarf Binaries
· *Advisor:* Dr. Marina Orio

Research & Technical Strengths

- **Space telescope data reduction and analysis**
  - Euclid NISP spectroscopy
  - JWST NIRCam, NIRSpec, MIRI imaging/spectroscopy
  - HST ACS/WFC and WFC3/IR imaging
- **Software development:** Python-based data reduction and analysis pipelines
- **Observation/survey planning:** HST, JWST, and Roman observation design

Observing Programs

- Co-I, Roman GAS 2001, *Roman eXtreme Deep Field (RXDF)* (PI: Haojing Yan)
  - Significant contribution to survey design/planning and observing strategy
  - Lead of RXDF Time-Domain Science Working Group
- Co-I in 6 approved JWST programs: PIDs 2883, 4557, 4695, 9969, 12503, 12774
- Member in JWST GTO *PEARLS* team (PI: Rogier Windhorst)

Grants

- **Roman Cycle 1 Program (GAS 2001)**: $40K ($1.2M total funding)

Awards

1. **Boain Dissertation Award**, University of Missouri – Columbia, 2026
2. **Dorina Kosztin Memorial Award**, University of Missouri – Columbia, 2026
3. **Thesis of Distinction**, University of Wisconsin – Madison, 2020

Publications

First-Author

1. *Galaxy Luminosity Functions at $z \sim 7-10$ Based on Public JWST Data.*
   B. Sun, & H. Yan (2026), MNRAS, submitted to MNRAS
2. *New Insight from the James Webb Space Telescope on Variable Active Galactic Nuclei.*
   B. Sun, H. Yan, W.-L. Tee, & F. Wang (2026), ApJ, 999, 152
3. *On The Very Bright Dropouts Selected Using the James Webb Space Telescope NIRCam Instrument.*
   B. Sun, & H. Yan (2025), ApJ, 978, 60
4. *A Mysteriously Tight Correlation of* Hα *and* [O III] *Line Fluxes Revealed by the Spectra from the James Webb Space Telescope NIRSpec Instrument.*
   B. Sun, & H. Yan (2025), arXiv preprint 2503.21896
5. *X-ray Spectra and Light Curves of Cooling Novae and a Nova-Like.*
   B. Sun, M. Orio et al. (2020), MNRAS, 499, 3006

Second/Third-Author

6. *GALEX + JWST: An investigation of star-forming galaxies in the low-z universe.*
   H. D.-Quiroz, **B. Sun**, & H. Yan (2026), in preparation
7. *A New Population of Point-like, Narrow-line Objects Revealed by the James Webb Space Telescope.*
   H. Yan, **B. Sun**, & R. Shive (2025), arXiv preprint 2509.12177
8. *PEARLS: 21 Transients Found in the Three-Epoch NIRCam Observations in the Continuous Viewing Zone of the James Webb Space Telescope.*
   H. Yan, **B. Sun**, Z. Ma et al. (2025), ApJ, 998, 115
9. *JWST view of four infant galaxies at $z = 8.31-8.49$ in the MACS0416 field and implications for reionization.*
   Z. Ma, **B. Sun**, C. Cheng et al. (2024), ApJ, 975, 87
10. *An Edge-on Regular Disk Galaxy at $z = 5.289$.*
    H. Yan, **B. Sun**, & C. Ling (2024), ApJ, 975, 44
11. *A Strongly Lensed Dusty Starburst of an Intrinsic Disk Morphology at Photometric Redshift of $z_{\rm ph} > 7$.*
    C. Ling, **B. Sun**, C. Cheng et al. (2024), ApJ*L*, 969, L28
12. *JWST's PEARLS: Improved Flux Calibration for NIRCam.*
    Z. Ma, H. Yan, **B. Sun** et al. (2024), PASP, 136, 024501
13. *JWST's PEARLS: Transients in the MACS J0416.1-2403 Field.*
    H. Yan, Z. Ma, **B. Sun** et al. (2023), ApJ*S*, 269, 43
14. *Candidate Galaxies at $z \geq 11.3-21.8$ and beyond: initial results from JWST's public data taken in its first year.*
    H. Yan, **B. Sun**, Z. Ma, & C. Ling (2023), arXiv preprint 2311.15121
15. *JWST's PEARLS: Mothra, a new kaiju star at $z = 2.091$ extremely magnified by MACS0416, and implications for dark matter models.*
    J. M. Diego, **B. Sun**, H. Yan et al. (2023), A&A, 679, A31

Others

16. *Constraints on the Pop III Sky Surface Brightness from High-Redshift Caustic Transits in MACS0416.*
J. Perivolotis et al. (2026) including **B. Sun**, submitted to ApJ
17. *A clumpy ring galaxy at $z = 4.0148$.*
D. Vizgan et al. (2026) including **B. Sun**, ApJ, 1004, 114
18. *JWST's PEARLS: Temperatures of Nine Highly Magnified Stars in a Galaxy at Redshift $z = 0.94$ and Simulated Stellar Population Dependence on Stellar Metallicity and the Initial Mass Function.*
H. Williams et al. (2026) including **B. Sun**, ApJ, 996, 105
19. *JWST's PEARLS: A candidate massive binary star system in a lensed galaxy at $z = 0.94$.*
H. Williams et al. (2025) including **B. Sun**, ApJ, 997, 292
20. *Constraining the $z \sim 1$ Initial Mass Function with* HST *and* JWST *Lensed Stars in MACS J0416.1-2403.*
S. Li et al. (2025) including **B. Sun**, ApJ, 988, 178
21. *Anatomy of a $z = 6$ Lyman-α emitter down to parsec scales: Extreme UV slopes, metal-poor regions and possibly leaking star clusters.*
M. Messa et al. (2025) including **B. Sun**, A&A, 694, A59
22. *Extreme Ionising Properties of Metal-Poor, $M_{\rm UV} \simeq -12$ Star Complex in the first Gyr.*
E. Venzella et al. (2024) including **B. Sun**, A&A, 691, A251
23. *NICER Monitoring of Supersoft X-ray Sources.*
M. Orio et al. (2022) including **B. Sun**, ApJ, 932, 45

Presentations

1. *New Insights into Galaxy Evolution: two discoveries with JWST data.*
Colloquium Talk at the University of Wisconsin-Madison, Madison, WI, 2026.5
2. *New Insights from JWST on Variable AGNs.* [1]
Oral (Special Session), 247$^{\rm th}$ AAS Meeting, Phoenix, AZ, 2026.1
3. *A New Population of Point-Like, Narrow-Line Objects Revealed by JWST.*
Press Conference, 247$^{\rm th}$ AAS Meeting, Phoenix, AZ, 2026.1
4. *A Mysteriously Tight* Hα-[OIII] *Correlation & Non-Case B Balmer Decrements Revealed by JWST/NIRSpec.*
Poster, 246$^{\rm th}$ AAS Meeting, Anchorage, AK, 2025.6
5. *Candidate Galaxies at $z \sim 7 - 10$ Based on Public JWST NIRCam Data in Cycle 1.*
Poster, 243$^{\rm rd}$ AAS Meeting, New Orleans, LA, 2024.1
6. *Investigating Environmentally Dependent Starburst Activities Using eROSITA and GALEX.*
Poster, 243$^{\rm rd}$ AAS Meeting, New Orleans, LA, 2024.1
7. *X-ray Spectra and Light Curves of Cooling Novae and a Nova-Like.*
Poster, 235$^{\rm th}$ AAS Meeting, Honolulu, HI, 2020.1
8. Five seminar presentations at the University of Missouri Astro Seminar

[1] Same topic presented at the 2025 Mid-American Regional Astrophysics Conference (MARAC), Columbia, MO, 2025.12.

Miscellaneous

1. **Teaching & Mentoring**
   - *Teaching*: taught four undergraduate-level physics and astronomy courses.
   - *Student Mentoring*: mentored undergraduate students in astronomical research, including training in data analysis and essential research skills.

2. **Outreach & Service**
   - *Public Outreach*: managed the Laws Observatory at the University of Missouri from 2022–2026, organizing public observing events and giving astronomy lectures to diverse audiences. Nearly 15 years of experience as a semi-professional astrophotographer, supporting astronomy outreach and public engagement.
   - *Departmental Service*: Familiar with the workflow and can contribute to seminar organization and other related departmental activities.